

Gout

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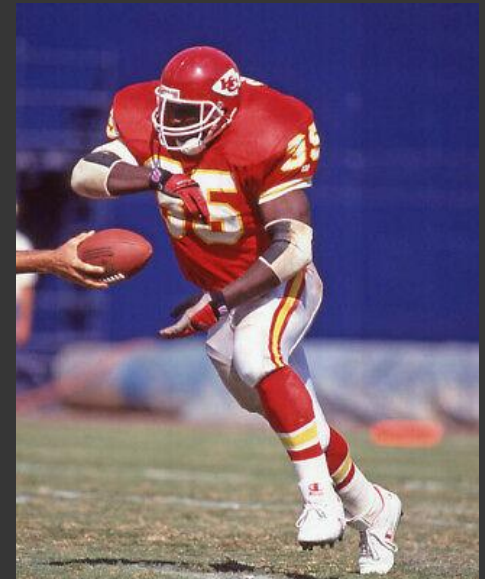
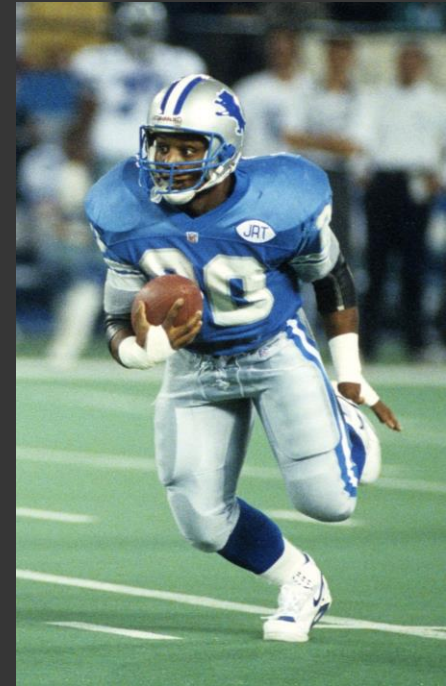
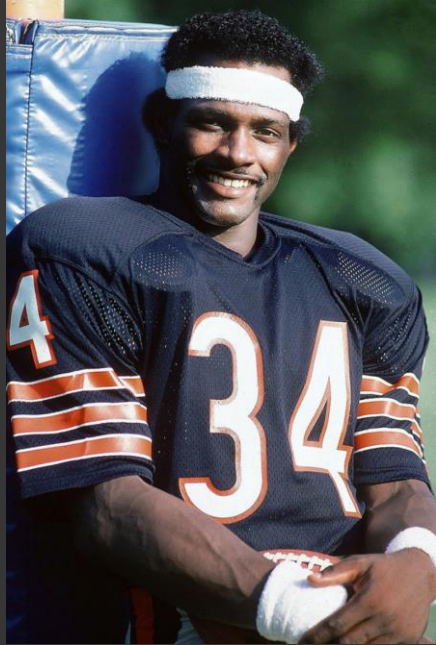
Assistant Clinical Professor

Loma Linda Univ School of Medicine

LECOM

- This record setting running back was told he had hyperuricemia in his early 30's while asymptomatic but later developed swollen toes at approximately age 42 and has battled this condition since then:

- Walter Payton
- Barry Sanders
- Christian Okoye
- Emmitt Smith
- Franco Harris



Case 1

- 75 yo man with 10 yrs of intermittent joint swelling in toes, ankles, knees; wrist of late
- PMH: Heart failure – uses furosemide QID
- 249 lbs, loves SPAM

Component Latest Ref Rng	1/17/2009
CREAT 0.7 - 1.3 MG/DL	1.5 (H)
GFR mL/min	49- NB
URIC 3.4 - 7.2 mg/dL	13.2 (H)



Key Learning Objectives

- Diagnosis, epidemiology
- Pathogenesis, crystal analysis
- Acute treatments
- Chronic therapies
- A “special” case

General Principles

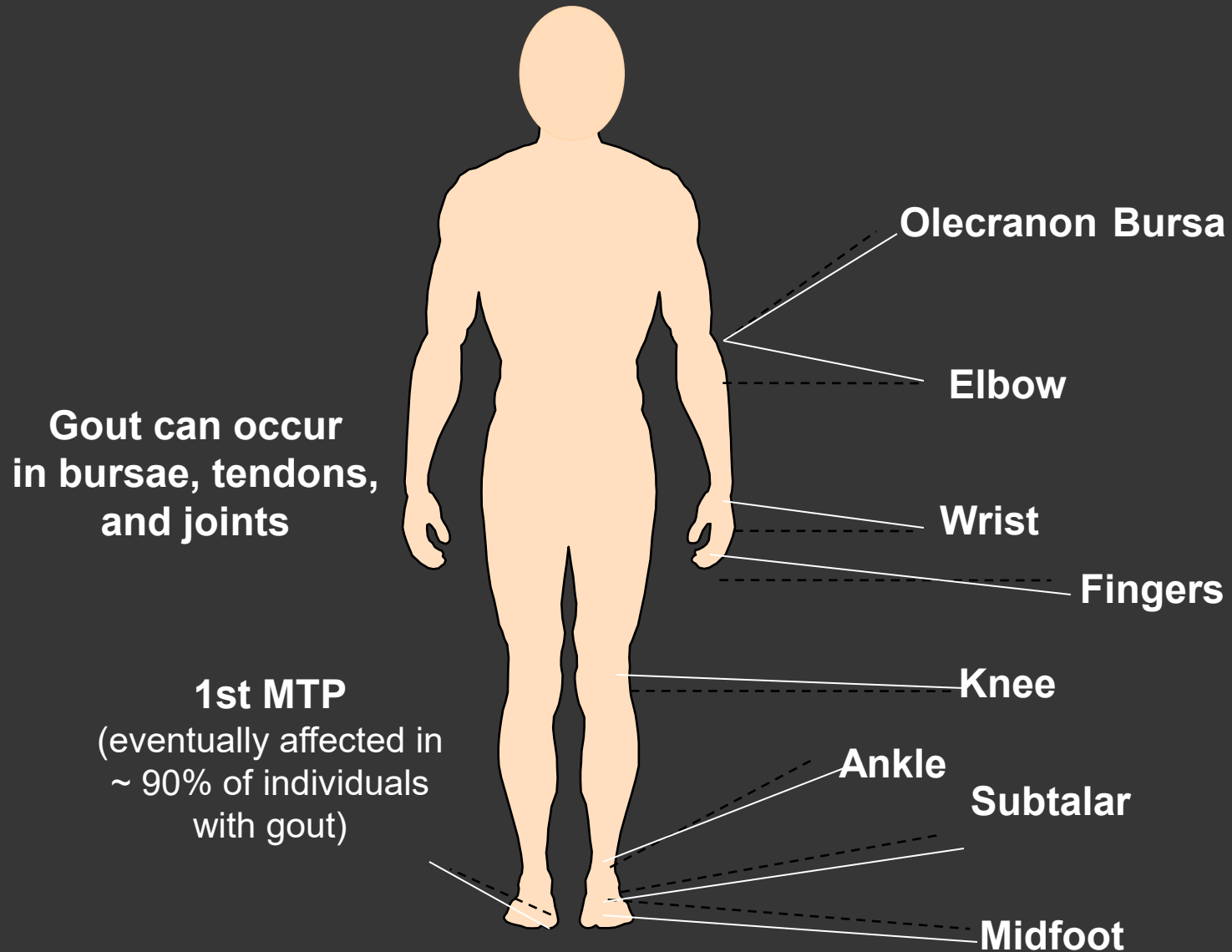
- Initially: Acute arthritis
 - Inflammatory response: monosodium urate (MSU) crystals
- Most common form of inflammatory arthritis in men
- Urate: end product of purine metabolism
- Blood level of urate > physiologic limit of solubility
(6.8mg/dl): Tissue crystallization

* Terkeltaub RA. N Eng J Med 2003; 349:1647-1655

Signs & Symptoms

- **acute monoarticular; distal lower extremities (50%)**
- **pain and tenderness, swelling, warmth and erythema; occasionally fevers**
- **frequently nocturnal**
- **attack may subside in 3-10 days without treatment**

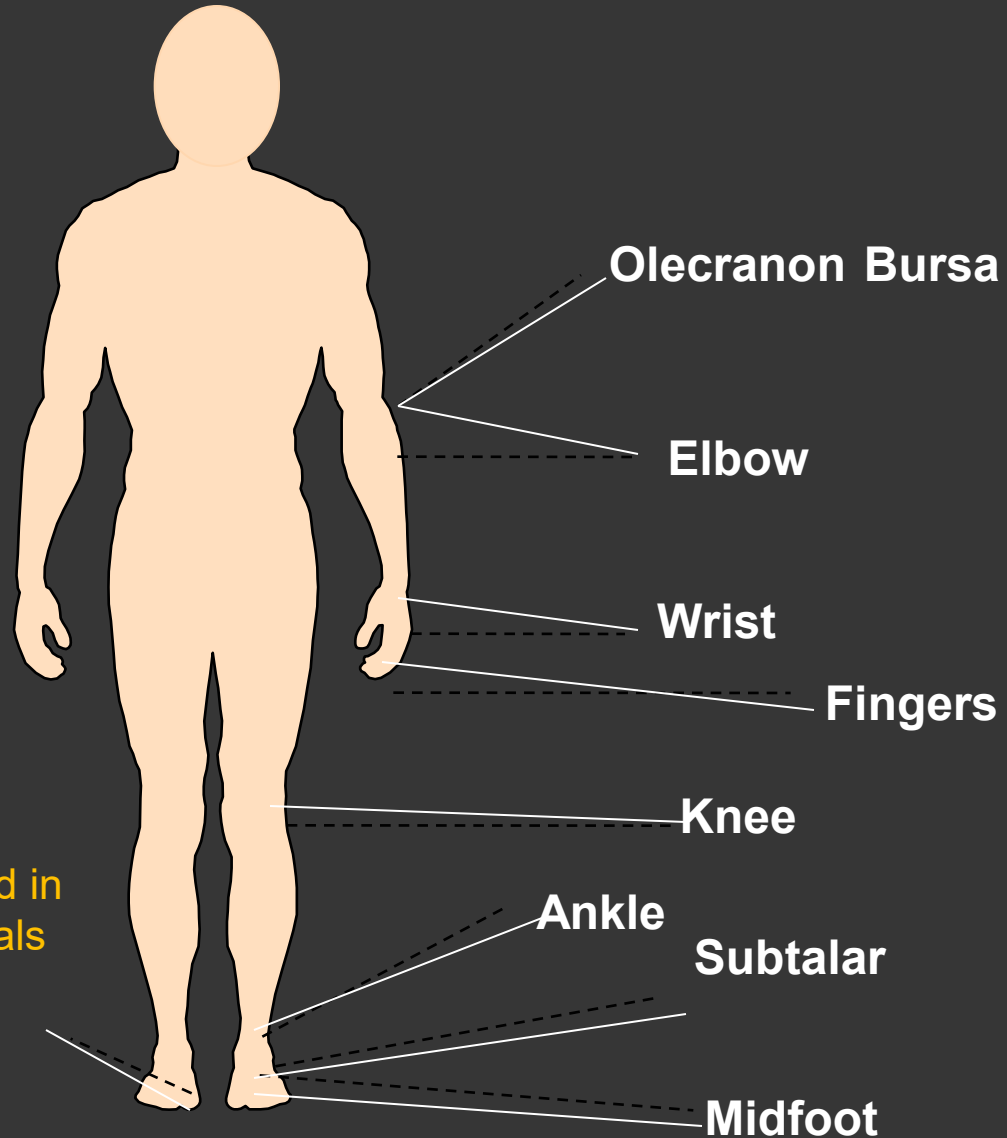
Common Sites of Acute Attacks



Common Sites of Acute Attacks

Gout can occur
in bursae, tendons,
and joints

1st MTP
(eventually affected in
~ 90% of individuals
with gout)



Diagnosis of Gout

- “Gold Standard” – demonstration of MSU crystals in PMNs in synovial fluid
- Classical history of attack(s) in typically involved joint with or without elevated serum uric acid
- Serum uric acid will be elevated at some point in time:
 - May be **normal or low** at time of attack
 - Retest several weeks after attack

Differential Diagnosis of Gout

- Exclusion of other possible diagnoses:
 - Septic joint (risk factors)
 - Reactive arthritis (antecedent infection – GU/GI)
 - Spondyloarthritis – young men, psoriasis
 - Trauma, hemarthrosis
 - Pseudogout (CPPD)
 - Hyperuricemia unrelated to diagnosis (RA, PsA, OA)

Evaluation

■ Clinical

- Family history
- Personal history
- Medications, diuretic use
- Diet, alcohol
- Metabolic Syndrome

■ Examination

- Evaluate all joints
 - tophi
 - bursae, tendons
 - ears

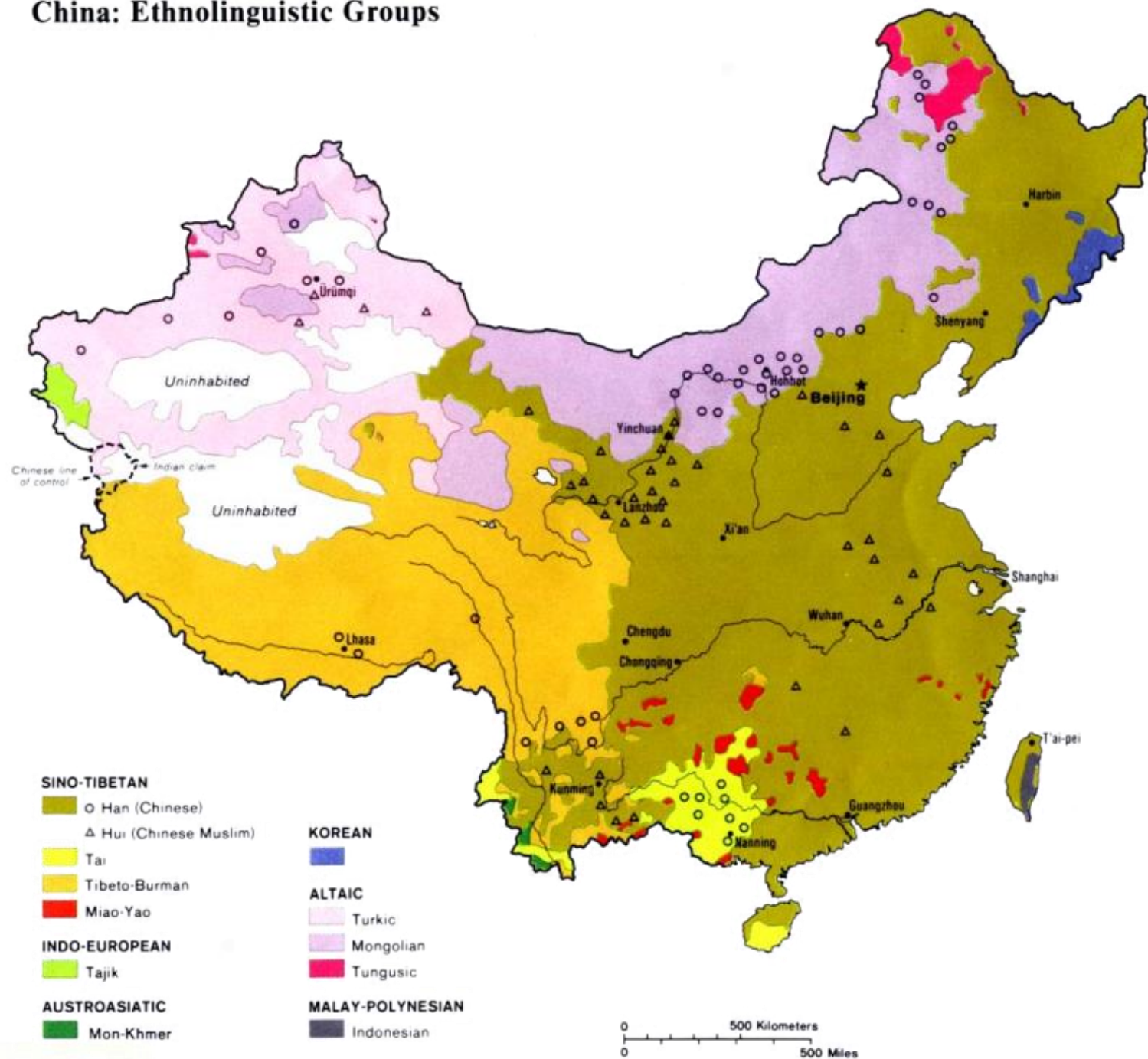
■ Radiographic

- Xrays
- US; others.....

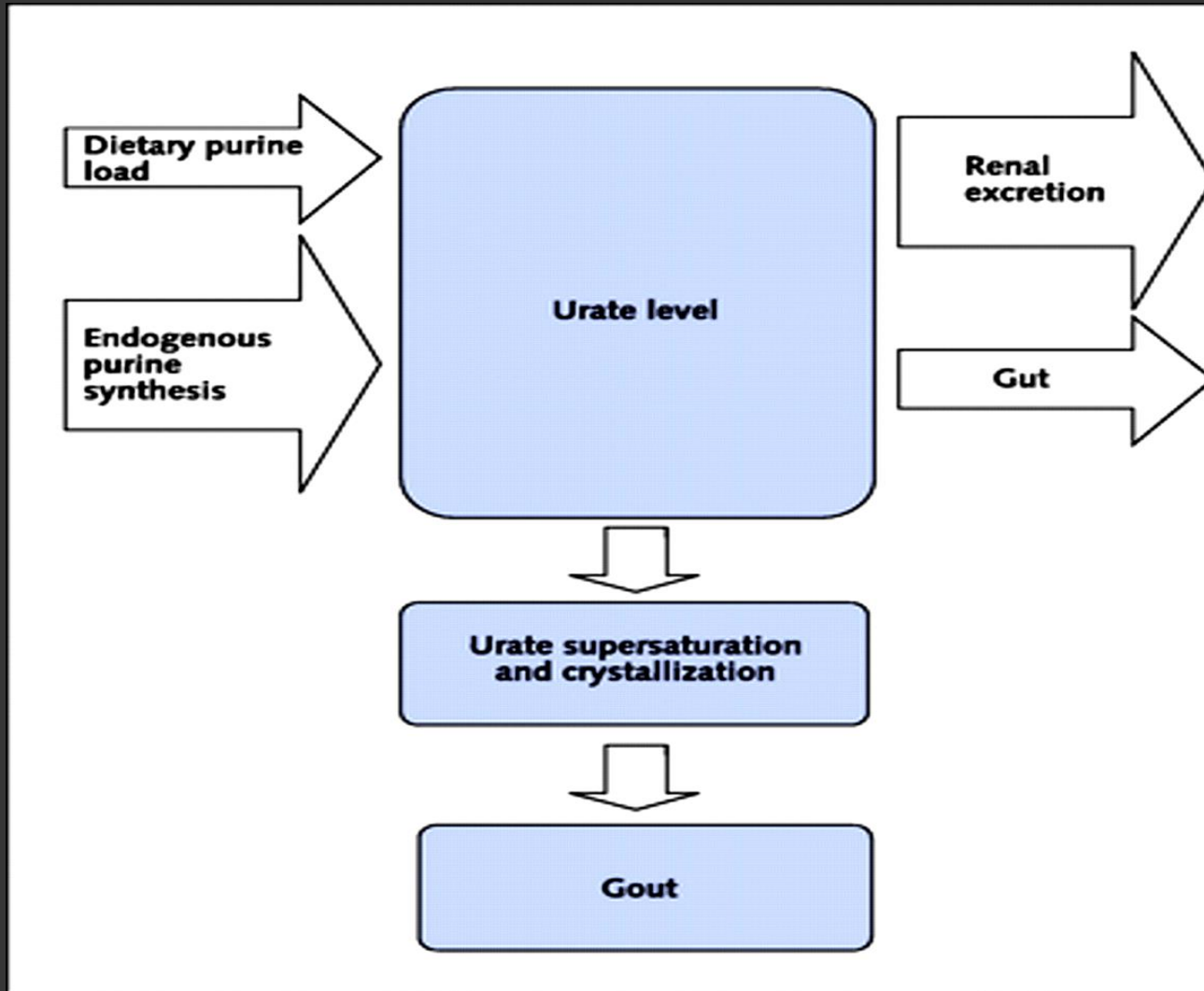
■ Laboratory

- CBC
- ESR and/or CRP
- Chemistry (LFT, Renal, uric acid)
- TSH
- HLA-B58 (Han Chinese, Korean)

China: Ethnolinguistic Groups



Overview of the Pathogenesis of Gout



Precipitating Factors

- Trauma/Surgery
- Low dose aspirin
- Diuretics-other medications
- Systemic illness

Risk Factors for the Development of Gout: Age and Gender

■ Men

- Higher serum urate levels (men > 7.0mg/dl, women >6.0mg/dl)
- In younger patients, gout overwhelmingly in men

■ Women

- Increased risk after menopause
 - ♦ ↓ estrogen → ↓ renal excretion of uric acid

Causes of Hyperuricemia

- **>90% Under-excretors:**
 - Reduced proximal tubular excretion of uric acid; genetics
 - Medication induced: Medications impairing renal urate clearance
 - **Renal insufficiency**
- **<10% Over-producers: de novo increased purine biosynthetic rate**
 - **>600mg/day**

* Scott JT, Pollard AC Ann Rheum Dis 1970;29:397-400

Medications Affecting Urate Excretion

- Thiazides and loop diuretics
- Low dose aspirin
- Cyclosporin A
- Anti-tuberculous medications:
 - pyrazinamide
 - ethambutal
- Niacin

Risk Factors for Development of Gout: Diet

- “Low purine diet” results in modest -1 mg - reduction in serum uric acid
- Risk from alcohol intake: Beer>liquor>wine



- High meat consumption
- High seafood consumption
- High **dairy** consumption
- High purine-rich vegetables

↑ risk of gout

↑ risk of gout

↓ **risk of gout**

no association

Secondary Causes of Hyperuricemia Due to Uric Acid Overproduction

- Myeloproliferative syndromes
- Psoriasis
- Tumor Lysis Syndrome
- Genetic Disorders
 - Deficient hypoxanthine-guanine phosphoribosyl transferase

Crystals

1



CPPD crystals

2



MSU crystals

- CPPD crystals are weakly positively birefringent, rhomboid, rods, squares, or irregular under compensated polarized light

1. <http://www.emedicine.com/med/topic1938.htm>.

2. ACR Clinical Slide Collection on the Rheumatic Diseases, 1998.

Treatment Options for Acute Attack?

- Nonsteroidal anti-inflammatory drugs
- Colchicine
- Corticosteroids
 - Intra-articular
 - Intramuscular, PO
- Interleukin – 1 antagonists?
 - Anakinra
 - Canakinumab
 - Rilonacept

Protection Against Further Attacks - Prophylaxis

Dosing

Complications of chronic use

Colchicine

0.6-1.8 mg daily

Dose adjusted for renal insufficiency

Neuromyopathy, rhabdomyolysis

Drug interactions with cyclosporine, statins, macrolides

NSAIDS

Any NSAID: lowest effective dose

NSAID gastropathy

Renal dysfunction

Cardiovascular; thrombotic

Treatment of Chronic Gout and Hyperuricemia

As we discussed there are many strategies for treating gout:

- 1. Flush your kidneys routinely - drink enough water to stay hydrated
- 2. reduce salt in your diet (to help protect your kidneys) and may help you need less water pills
- 3. Reduce your portions and calories to achieve a healthier weight
- 4. Eat any vegetable you want
- 5. Reduce red meat and seafood and animal protein; beer and liquor
- 6. Fruits are ok, but sugary foods make gout worse
- 7. Take a vitamin C supplement - it helps gout
- 8. increase low or nonfat dairy

- 9. Use colchicine once daily or twice daily if tolerated - avoid grapefruits and grapefruit juice while on colchicine
- 10. We will slowly add low doses of allopurinol - you may experience a flare of gout but don't stop this
- 11. Periodically we will measure your uric acid and adjust your medicines.

- Our patient:
 - Furosemide 20 QD
 - Low salt

⛶ Component Latest Ref Rng	1/17/2009
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+

Component <i>Latest Ref Rng</i>	1/17/2009	3/13/2009
CREAT <i>0.7 - 1.3 MG/DL</i>	1.5 (H)	1.3
GFR <i>mL/min</i>	49-NB	58-NB
URIC <i>3.4 - 7.2 mg/dL</i>	13.2 (H)	8.5 (H)

- Our patient:
 - Furosemide 20 QD
 - Low salt
 - Allopurinol 100 mg qd

+

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- Our patient:
 - Furosemide 20 QD
 - Low salt
 - Allopurinol 100 mg qd

Component Latest Ref Rng	1/17/2009	3/13/2009	5/22/2009
CREAT 0.7 - 1.3 MG/DL	1.5 (H)	1.3	1.0
GFR mL/min	49-NB	58-NB	78-NB
URIC 3.4 - 7.2 mg/dL	13.2 (H)	8.5 (H)	6.0

Urate Lowering Treatments

■ Urostatic agents

- Xanthine oxidase inhibitors
 - **Allopurinol**
 - **Febuxostat/Uloric (2009)**

■ Enzymatic-pegloticase

- Krystexxa (2011)

■ Uricosuric Agents

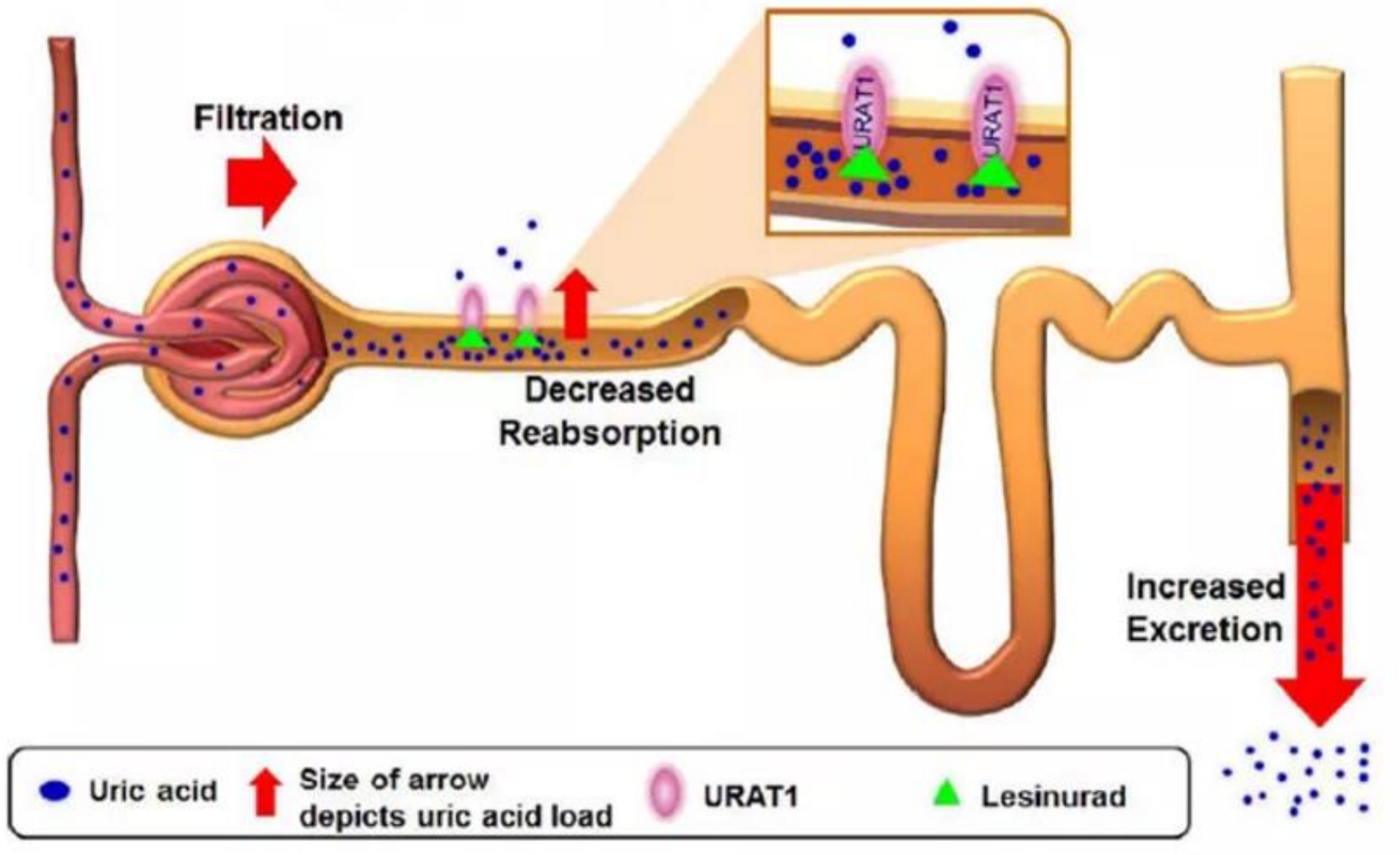
- Probenecid
- Losartan
- Fenofibrate

■ Lesinurad (URAT1 inhibitor)

- Removed from market 2019

Renal excretion of uric acid

Schematic of Lesinurad's Mechanism of Action



Treatment of Chronic Gout

- Allopurinol: Uricostatic drug of choice
 - Blocks the conversion of hypoxanthine to xanthine to uric acid
 - Commonly under-dosed
 - Measure serum uric acid every 3 to 6 months
 - Monitor for toxicity
 - Goal: Serum urate below 6mg/dL; 5mg/dL if tophi



Case 2....

- 44 year old HM presents with 2 months of worsening swelling in L big toe at base. No redness, pain initially, but now tight and even with some skin breakdown and oozing.
- PMH: hypothyroidism
- SH: prior heavy beer use, none in 2 yrs



- What lab is most likely to reveal the best target for treatment?
 - 24 hour urine uric acid
 - Serum phosphorus
 - Serum interleukin 6 levels
 - TSH
 - hypoxanthine-guanine phosphoribosyltransferase

Advanced Chronic Gout: Clinically Apparent Tophi



Advanced Gout: Late Radiographic Changes

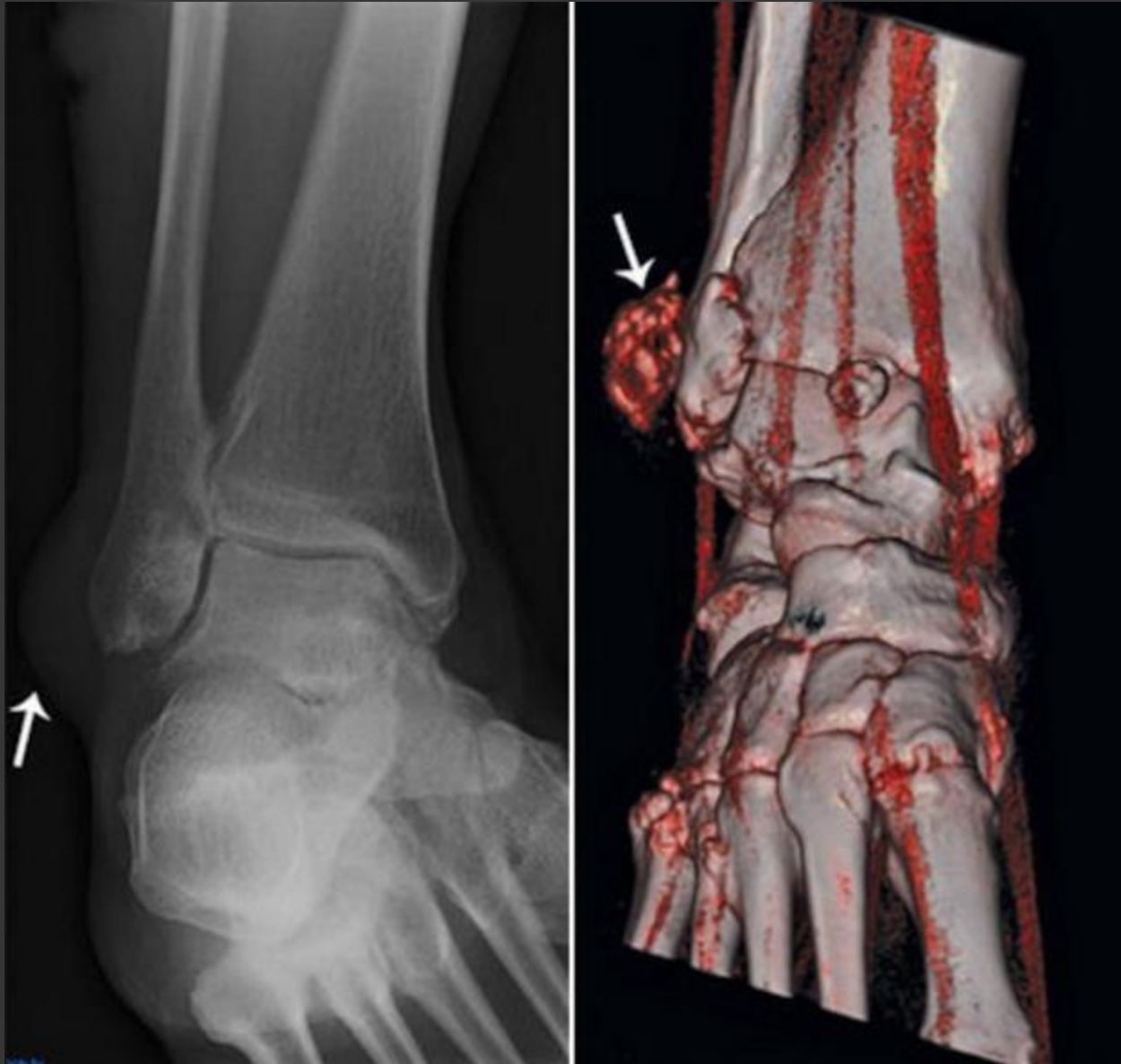
- destructive and hypertrophic; “overhanging edges”
- joint space is often preserved until late in disease process



Earlier diagnosis?

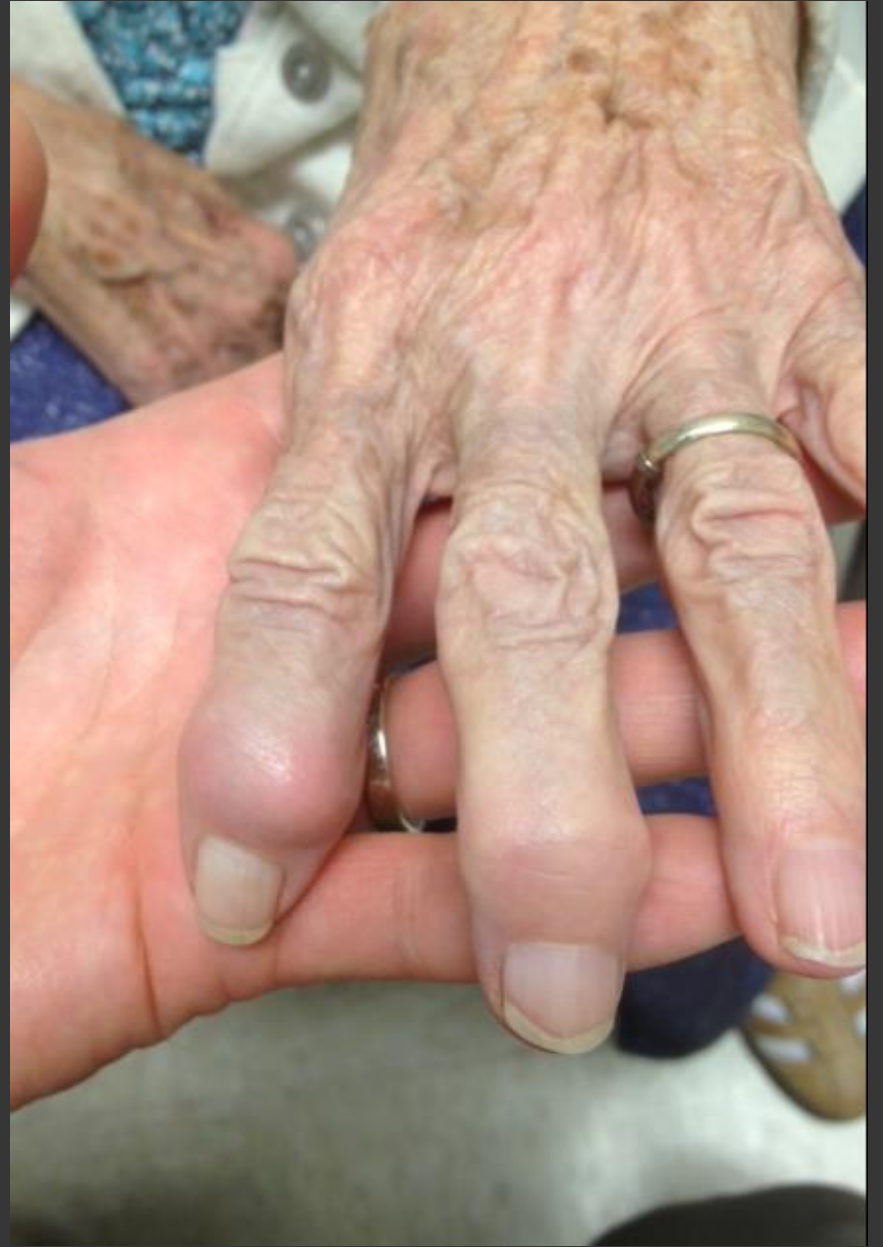


Dual Energy CT (DECT)



One Final Case.....

- CC: finger pain, swelling
- HPI: 84 yo WF with known OA complains of chronic aching and swelling of DIPs – gradually over past year; episodes of warmth and tenderness that comes and goes, but no pain in feet, knees or swelling elsewhere.
- PMH: hypertension, DMII, no psoriasis, crohns
- Meds:
 - hydrochlorothiazide 25 mg daily
 - metformin 500 mg bid
- Exam:
 - Moderate heberden and bouchard nodes, moderate mal-alignment, but with no MCP synovitis
 - soft tissue swelling in DIPs of L hand



What is the best explanation for these symptoms?

1. Erosive osteoarthritis
2. Elderly onset gout
3. Calcium pyrophosphate deposition disease
4. Rheumatoid arthritis
5. Multicentric reticulohistiocytosis

Elderly Onset Gout (EOG)

- subacute/chronic polyarticular hand involvement
- localization of tophi on heberden nodes
- increased female/male ratio
- association with drugs → decrease renal urate excretion (diuretics and low-dose aspirin)

In Summary:

- Traditional and newer RFs for gout
- Pathogenesis of gout – SUA metabolism; systemic disease
- LOW DOSE ALLOPURINOL
- GRADUAL TITRATION
- Treat to target
- Newer therapies



TINY CRYSTALS.
WIDESPREAD THREAT.

Uncontrolled gout has systemic consequences.

Uncontrolled gout can spread beyond the joints to almost anywhere throughout the body. MSU crystals can lodge in vital organs, cause severe disease, exacerbate comorbidities, and severely impact patients' lives.